THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 11: **Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the Earth stopped rotating, the part facing the Sun would: 1

(a) have rain **(b)** have continuous daylight
(c) have night **(d)** cool down

2. The Moon's phases when seen from the northern hemisphere: waxing crescent appears in the: 1

(a) morning sky only **(b)** evening sky (west)
(c) noon **(d)** midnight

3. During a total solar eclipse, the observer is in the Moon's: 1

(a) shadow of Earth **(b)** penumbra **(c)** sunlight **(d)** umbra

4. If the Earth's axis were not tilted, we would have: 1

(a) no seasons **(b)** no day **(c)** more seasons **(d)** no night

5. A full moon rises at about: 1

(a) sunrise **(b)** sunset **(c)** midnight **(d)** noon

6. In a partial eclipse, the observer is in the: 1

(a) total darkness **(b)** full sunlight **(c)** penumbra **(d)** umbra

7. In June, which hemisphere has long days? 1

(a) northern **(b)** southern **(c)** both equally **(d)** neither

8. A new moon rises at about: 1

(a) sunset **(b)** sunrise **(c)** midnight **(d)** 3 p.m.

9. A lunar eclipse lasts longer than a solar eclipse because: 1

(a) the Moon is large **(b)** the Sun is small
(c) the Earth's shadow is large **(d)** the Moon moves fast

10. In December, the southern hemisphere has: 1

(a) winter **(b)** the same as north
(c) summer **(d)** spring only

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do days and nights have unequal lengths in different seasons? 2

12. Why are the Moon's craters preserved for a long time? 2

13. Explain why the temperature is lower in winter. 2

14. Why do places near the equator have almost no seasonal change? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain with a diagram in words how a total solar eclipse is formed. 3

16. Describe the phenomenon of midnight Sun. 3

17. Why do astronauts jump higher on the Moon? 3

OR

Explain how the Moon's phases can be shown with a ball and a lamp. 3

18. Explain how seasons affect agriculture in India. 3

OR

Why is the sky dark during a total solar eclipse? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Draw (describe) the positions of the Earth in its orbit at the two solstices and the two equinoxes and write the seasons in both hemispheres. 5

OR

Explain why the Earth is slightly flattened at the poles (qualitative) and the effect of rotation. 5

20. Explain with a labelled description how the phases of the Moon are caused and name eight phases in order. 5

OR

Discuss the importance of the Moon for Earth: night light, calendars, festivals and study of space. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND